

Glossary

Pincushion and Barrel distortion: This is the geometric distortion in monitors where a geometric shape doesn't appear as it should. Pincushion distortion is when the screen bends inwards from the edges, whereas barrel distortion is when the screen bends outwards. This can be corrected by the special circuitry in the monitor.

Screen regulation: The screen expands in size when it suddenly turns bright from dark. This could be either in some part of the screen or the complete screen size might change.

Focus matrix: This is the pattern that tells how good the sharpness of the monitor is. It produces a coarse and fine test pattern on the screen. Many monitors show the pattern sharp in the middle and blurred around the corners.

Streaking and ghosting: When there is substantial difference in the intensity in adjacent areas of the screen, the streaking and ghosting effect occurs. The faint but visible trail (like a comet) visible in low intensity areas at the edge of the bright area is known as streaking. Ghosting on the other hand is the shadow formed at the edge of the image.

Black- and white-level shift: The shifting of the white or black colour on to the adjacent colour is known as level shift. The edge here will appear blurred.

RGB colour purity: This test produces pure RGB colour one by one on the screen and any irregularities are looked out for. Monitors mainly show more than one shade of a colour on the screen, whereas in the ideal case it should produce the colour precisely and uniformly across the screen.

Defocusing, blooming and halos: The bright image will always look thicker than the image less bright. This is a function of both the CRT and the human visual system.

CRT Shadow Mask Distortion: When the screen is kept bright for a long period of time, the ghost image of that screen will stay as a faint image on the screen even after the screen is changed. This is Shadow Mask Distortion.

TEST PROCESS

The monitors we tested were evaluated on the basis of two main parameters: performance and features.

While testing for performance, we used DisplayMate as the benchmarking tool. This software generates test images using which we examined the features of the monitors. We checked for the clarity of patterns displayed on the screen, the accuracy of the screen's internal geometry (whether the lines were parallel or not), the effect of screen brightness on the size of an image, the monitor's ability to accurately display an image at high

resolution, and whether there was a streaking effect when there was a change in intensity in one part of the screen.

We also looked for white and black level shift, colour purity, screen uniformity and flickering.

Among the features that we looked for were, data memory for video modes, number of storable modes, clarity of the OSD text and energy saving features. Apart from these, we also checked out factors such as ease of use, help and troubleshooting features, build quality and price.

distortions such as lines not being parallel, circles appearing oval and squares looking like rectangles. 15-inch monitors using the Shadow Mask technology witness such geometric distortions due to the fact that if the dots or pixels on the shadow mask are very bright, they tend to bleed together making lines look thicker or jagged. The distortion could be due to the inaccuracy of the electron beam deflection or faults in the support electronics. The distortions are usually prominent at the corners of the screen.

The DisplayMate test for geometric distortion and screen regulation revealed that almost all the monitors in the 15-inch category had visible distortion except for Sony CPD-E100 and ViewSonic E53. But even these monitors were not able to give a distortion-free reading when we checked on the left and right edges using a digital Vernier calliper.

However, the 19-inch monitors performed quite well with hardly any increase in screen size except in the case of Samsung SyncMaster 900NF which gave a low score in the geometry test since the lines were not perfectly parallel and there was a slight expansion in the screen when the brightness was increased. Acer's P911 monitor did well in this test, though the 15-inch monitor from Acer returned unsatisfactory results.

In the 21-inch category, the LG

StudioWorks 221U was almost perfect in producing parallel lines and gave above average result in the screen regulation test, especially in the symmetric and asymmetric regulation test.

The geometry of the monitor is important for those who are using CAD software, especially on a 19-inch or a 21-inch monitor, since these are the people who have to deal with lines, circles and polygons all the time. So, if the monitor is not capable of generating these perfectly, the design that appears on screen may not be up to the mark.

There are many free utilities such as vgatest available on the Net, which you can use to minimise the geometric distortion. This software, for instance, produces a test pattern on the screen, using which you can adjust the pincushion and barrel distortion.

Sharpness and resolution

test: These tests are important for those in design-related professions. Sharpness and resolution is what you pay for and if, after creating what you think is a perfect image it doesn't show up that well, then most likely the monitor would be the culprit.

For testing the image sharpness and clarity of resolution, we ran seven different patterns in DisplayMate. The first pattern was the horizontal

bar resolution test where there were a total of five bands with the vertical bar at the



The 19-inch Acer P911 has a good build quality and reproduces sharp images